

# The Blob in Mirror — Canonical Armature V347.1

Blob Principle — Master V347 (English translation)

Rudolphe Mirante (ORCID 0009-0001-0990-7122)

French original: 9 September 2026 · English translation: 6 October 2026

**Translation note.** This is an AI-assisted English translation (Claude, Anthropic) of the French original *Le Blob en miroir — Armature canonique V347.1*, made on 6 October 2026 with the project’s locked lexicon (index/LEXIQUE\_FR-EN.md). It has **not yet been reviewed by the author**. The French original is the authoritative text. Corrections are welcome as GitHub issues (challenge DEFI-L2).

Two passages are corrupted in the French source PDF itself (p. 15, section II.8 opening, and p. 17, RO-21 entry, where text segments appear spliced). They are translated literally, without reordering. A few glyphs missing in the source (Thm 7.3, Hyp. 8.1, Def. 9.1, Target 11.13) were reconstructed from context and should be checked against the French.

**Status block — TOE demonstrated: NO.** Formalised speculative research programme, candidate for external evaluation. Gates G0–G5 open, G6 external, physical gates 0/7. Corpus DOI: 10.5281/zenodo.23188586. Licence: CC BY-NC-ND 4.0.

## BLOB PRINCIPLE — MASTER V347

### THE BLOB IN MIRROR

### CANONICAL ARMATURE V347.1

Rudolphe Mirante — Hâpy-Com SASU

9 September 2026 — Europe/Paris

### CONTENTS OF THE ARMATURE

V347.1 is organized as a causal mirror: narrative → formal object → residue → open research.

|    |                                                                |
|----|----------------------------------------------------------------|
| 0  | Status, rules, mirror contract and genealogy of the parameters |
| I  | Book I — The cosmogony of the Blob: I.1 to I.13                |
| II | Book II — The same story, in mathematics: II.0 to II.13        |

|     |                                                      |
|-----|------------------------------------------------------|
| III | Book III — Register of the 27 open research problems |
| P   | Sifting plan from V346 to the complete MASTER V347   |
| G   | Minimal glossary                                     |
| J   | Log of the V347.1 corrections                        |
| V   | Provenance, versioning and freeze                    |

**TOE demonstrated: NO • G0-G5 open • G6 external • Physical gates: 0/7**

# Mandatory status block

**TOE demonstrated: NO.** Scientific status: formalized speculative research program, candidate for external evaluation. **G0-G5: OPEN. G6: EXTERNAL. Physical gates: 0/7.**

Any citation of V347 that omits this block distorts the program.

V347.1 is neither a declaration of victory nor an academic reduction of the Blob. It is its falsifiable causal specification: the intuition in its order of birth, its actual mathematical twin, then the gap between the two transformed into named work.

# Rule of V347

**State of the present document.** V347.1 is the audited canonical armature. It fixes the causal order, the statuses, the residues, the open research problems, the genealogy of the parameters and the sifting protocol. It does not yet claim to contain the 600–800 pages of V346 material that must be transcribed in extenso into the complete Books I-III. The armature may evolve under V347.x; the name MASTER V347 will be frozen only after integration of the complete corpus and hashing of Books I and II. After this freeze, any substantial modification will create V348.

**Principle.** V347 adds no physical law to the Blob. It may, on the other hand, derive logical consequences of the definitions already present, identify a contradiction, retract an impossible formulation and name the missing link. A correction is not a new law.

**Book I — the compass.** It narrates the consolidated cosmogony as it appears in the V346 provenance: substrate, scission, twin lineages, simplicial carrier, mitosis, three times, two irreversibilities, edge, triangle, tetrahedron, opposition, bath, chirality, viability, construction. It contains no mathematical formalism and claims no proof status. It does not claim that every narrative formulation historically preceded all knowledge of physics; its chronology is attested by the provenance references.

**Book II — the mirror.** It takes up the same sequence with only the objects actually constructed. Internal historical terms such as  $t_0$ ,  $t_1$ ,  $t_2$  or “dissipative” may appear in it when they are formally defined; on the other hand, no identification with an observed entity — photon, graviton, physical matter, fundamental interaction, Einstein, Maxwell, Standard Model,  $\Lambda$ CDM — is admitted in a formal statement. These identifications are deferred.

**Book III — the residue.** It consolidates all the open research problems. Each RO has either a falsifier, or a closure criterion, or an explicit claim bound. A status downgrade is a result in the same right as a promotion.

## Mirror contract

Each pair I.k / II.k must answer five questions, in this order:

1. **Intuition** — what exactly does the narrative assert?
2. **Object** — what mathematical object actually exists?
3. **Proof** — what is its status: axiom, definition, theorem, certificate, measurement, development, failure?
4. **Residue** — CARRIED [PORTÉ], WEAKENED [AFFAIBLI], CORRECTED [CORRIGÉ] or ABSENT [ABSENT]?
5. **Closure** — what result would make this residue evolve, and what would kill it?

## Reading rules

1. **Priority of statuses.** The SUPERSEDED\_BY rule of Core 1.0 dominates any old V346 label. [ARCH] means: result limited to a historical model, not transported to the canonical sector.
2. **No resurrection.** [R] and [FAIL] may be narrated as provenance; they do not become true again through repetition.
3. **No new feedback from reality.** The freeze is retrospective, not a historical preregistration. V347 guarantees only that no new physical comparison can be used to retroactively fill a hole in Books I-II. The past remains visible, including its influences and its errors.
4. **Dependency discipline.** A symbol must be defined before use or carry an explicit reference to its definition. II.0 is a map and may make forward references; theorems, for their part, must declare all their hypotheses.
5. **Genealogy of the parameters.** Every numeric constant receives, independently of its epistemic status, a provenance: DERIVED [DÉRIVÉ], DEFINITIONAL [DÉFINITIONNEL], PRODUCTION, ARCHIVE, FREE/OPEN [LIBRE/OUVERT]. A production value is never a fundamental constant.

## Epistemic statuses

[T] theorem; [CERT] finite certificate; [M] measurement; [DEV] development; [H] hypothesis; [OPEN] open obligation; [FAIL] failure; [R] retracted; [EXT] external validation; [ARCH] result of an archived model; [A] axiom/posited variable; [DEF] definition; [METH] methodological rule; [R-ALG] inherited algebraic error corrected in V347.

## Minimal genealogy of the parameters

| Parameter      | Layer   | V347 provenance      | Scope                           |
|----------------|---------|----------------------|---------------------------------|
| $\beta = 0.30$ | A_Gibbs | ARCHIVE / PRODUCTION | historical weight of D in S_sim |

| Parameter                                 | Layer          | V347 provenance       | Scope                                              |
|-------------------------------------------|----------------|-----------------------|----------------------------------------------------|
| $\tau = 0.50$                             | A_Gibbs        | ARCHIVE / PRODUCTION  | historical sampler temperature                     |
| $\kappa_0 = 8$                            | A_Gibbs/Rips   | ARCHIVE / PRODUCTION  | historical substrate density/radius                |
| $d\star = 2\pi/\arccos(1/3)$              | B              | DERIVED               | target geometric coordination                      |
| $\alpha = 0.4$ ; $b = 0.05$ ; $\mu = 3.0$ | B              | PRODUCTION            | G0 campaign; not fundamental                       |
| $g = 1.5$                                 | A, twin sector | ARCHIVE / FREE        | lineage-grading coupling, not derived              |
| $\kappa$ (MARKED weight)                  | B, analysis    | DEFINITIONAL/ANALYSIS | norm parameter; $\kappa=1$ in certain measurements |

## Correspondence table of the two books

| Ch. | Book I — cosmogony                 | Book II — formal object                                                                                    | V347.1 residue                                                   |
|-----|------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1   | The substrate that is not nothing  | $\Omega^0$ , initial object, Dirac measure; domain of $S_{\sim}$ at the frontier                           | <b>CARRIED structure / WEAKENED potentiality</b>                 |
| 2   | Scission and twin lineages         | $\sigma, \sigma_0$ ; archived domains; global twin space not constructed                                   | <b>WEAKENED</b>                                                  |
| 3   | Simplicial carrier and mitosis     | $\mu, t_1$ ; only sectoral conservation; reversible $\mu$ ; nucleation open                                | <b>WEAKENED eternity / CARRIED genealogy / ABSENT nucleation</b> |
| 4   | Three times, two thermodynamics    | $t_0, t_1, t_2$ ; KL-Lyapunov; $t_2$ is sampler; absence of dissipation at $\Omega^1/\Omega^2$ not general | <b>CORRECTED</b>                                                 |
| 5   | Point, edge, triangle, tetrahedron | $\Delta^k$ ; $\mu$ -only no-go; $\partial\Delta^2 \neq \Delta^2$ ; closure not formalized                  | <b>CORRECTED / ABSENT closure causality</b>                      |

| Ch. | Book I — cosmogony   | Book II — formal object                                                                    | V347.1 residue                               |
|-----|----------------------|--------------------------------------------------------------------------------------------|----------------------------------------------|
| 6   | Birth of matter      | first 3-simplex; $T^3$ sector; action $S(T)$ ; exact dissipation at $\Omega^3$ not general | <b>CARRIED volume / WEAKENED dissipation</b> |
| 7   | Inherited opposition | involution, Dirac-Kähler, $\sigma$ domains [ARCH]; two global universes absent             | <b>WEAKENED [ARCH]</b>                       |

| Ch. | Book I — cosmogony                          | Book II — formal object                                                        | V347.1 residue                                                |
|-----|---------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------|
| 8   | The bath                                    | sketches $\text{By}, \eta$ ; no complete operation $\mathcal{R}$               | <b>ABSENT</b>                                                 |
| 9   | Chirality                                   | mirror invariance, $E[\text{O}_{\text{net}}]=0$ , correlation without net hand | <b>CORRECTED</b>                                              |
| 10  | Everything appears, not everything persists | local exploration, Gibbs, C2-R; accessibility and persistence open             | <b>WEAKENED</b>                                               |
| 11  | Construction and continuous victory         | R-TRUE, quotient, pull-back, MARKED, GEN-4                                     | <b>CARRIED transmission / ABSENT generational improvement</b> |
| 12  | Resulting world                             | deferred confrontation contract                                                | <b>ABSENT by construction</b>                                 |
| 13  | The why                                     | outside the formalism                                                          | —                                                             |

The four residues are frozen: CARRIED = the formalism says what the narrative says; WEAKENED = it says less, or only under hypotheses/window; CORRECTED = the formalism contradicts part of the narrative and the narrative yields; ABSENT = no corresponding object exists yet.

# BOOK I — THE COSMOGONY OF THE BLOB

*Register: narrative. No statement in this book has the status of proof. The references serve provenance.*

## I.1 — The substrate that is not nothing

Before there is a space, before there is a thing, before there is a before, there is a support. This support has no extension, no duration, no direction. It is not born, does not age, does not disappear. It is not the void, because the void is already something that is missing from something; it is that on which a thing can appear. We call it the substrate, and we give it, in the language of the corpus, the name patient zero: that which is there before the first simplex, in dimension zero (V346 p. 1028). In the cosmogony, the substrate is said to be cold: it has nothing to wear out and no thermodynamics is yet defined in it. This sentence is narrative; Book II will verify separately whether a corresponding formal quantity already exists at  $\Omega^0$ . It is eternal, not in the sense of an infinite duration, but in the sense of an absence of time: there is in it neither before nor after (V346 p. 128, 547). It contains no structure and, for this exact reason, no asymmetry: everything that will later be able to distinguish a left from a right, a past from a future, a thing from its opposite, will have to appear after it, never within it. One

question remains open, and the narrative does not claim to answer it: what makes the substrate set itself in motion? The corpus marked it as a foundational question from the origin (V346 p. 176, Q-2.4). The narrative begins at the first motion, not before.

## **1.2 — The first scission and the two twin lineages**

The first event is not the appearance of a thing. It is the appearance of an opposition. The substrate splits. But it does not split into two identical copies: it splits into two lineages of opposite polarity. From the same act are born simultaneously a branch and its opposite, a side and its mirror. In the author's intuition, these are two opposite simplices that are born together from the substrate, and with them two twin universes, each of which is the inverse of the other (V346 p. 1028, 1077). The initial pair is exactly neutral: what one carries, the other carries in negative. But the neutrality of the whole, at every scale, is not a consequence of heredity alone: if one lineage divides more often than the other, the sum is no longer zero. Global neutrality remains exact only if the two universes evolve as mirrors of one another. That, precisely, is the condition that the narrative assumes and that the formalism will have to supply. Each of the two universes, taken separately, is asymmetric; orientation is not created, it is shared between two twins. One thing must be said here, because the narrative must remain honest towards what mathematics later established: the direction of the scission, which twin receives which polarity, is not explained by the narrative. It is a choice made once, at the first division, and then transmitted without ever being remade. The corpus reduced it to exactly that: one bit set at the first division, everything else being inherited (V346 p. 1077).

## **1.3 — The simplicial carrier and mitosis**

The carrier, that is to say the primordial 0-simplex born of the scission, the vertex itself, belongs to the eternity of the substrate. It is not subject to wear. What is inscribed in it is not erased. This must be said precisely, because the word "simplex" is ambiguous: the edges, the triangles, the tetrahedra that the carrier will later construct are made and unmade; it is the carrier, the vertex, that does not die. The carrier possesses a single capacity: to divide. We call this division simplicial mitosis. One point engenders another, and the edge connecting them is the trace of this engendering. Each mitosis is an engraved act: because the carrier is indestructible, the act of its division remains inscribed forever in the genealogy. Mitosis wears nothing out, loses nothing, heats nothing. It adds. And it transmits. Each child receives the lineage of its parent. The polarity born at the first scission thus descends from mitosis to mitosis, without anything in the subsequent dynamics being able to rewrite it, because the subsequent dynamics acts on what is constructed, never on the act of birth (V346 p. 1028, §96.1).

## **1.4 — The three times and the two thermodynamics**

There is not one time. There are three, and they do not appear together. Time 1 is eternity. It is the time of the substrate and of the carrier. It has no arrow, no loss, no before or after. It is not a time that flows; it is a time that is. Time 2 is direction. It is born with mitosis. As soon as a carrier has divided, there exists a parent and a child, and this order cannot be inverted: what has been engraved remains engraved. It is an irreversibility without wear. Nothing is lost, but something accumulates. This is the first thermodynamics of the Blob: a thermodynamics of inscription, carried by the immortality

of the substrate (V346 p. 128, V183). Time 3 is dissipation. It appears only with matter. As soon as there exists an inside, something can transform, lose, heat up, come undone. This is irreversibility through wear, the only one we feel. This is the second thermodynamics of the Blob: a thermodynamics of friction, which dissipates (V346 p. 686, 1204). Two irreversibilities of opposite origins, then: the one because nothing is erased, the other because everything wears out. In the corpus, these three times will be found under the technical names  $t_0$ ,  $t_1$ ,  $t_2$  (eternity, mitosis, dissipation), and the two thermodynamics under the names type 1 (accumulation) and type 2 (dissipation).

## **1.5 — Point, edge, triangle, tetrahedron**

Mitosis does not construct everything at once. It makes levels appear, and each level opens a possibility that the previous one did not have. The point is the possibility of existing. Alone, it can do nothing. The edge is the possibility of relation. Two existences become connected. It is the first structure, and it is still reversible: it can be undone without losing anything. The triangle is the possibility of closure. Three mitoses that do not extend into a chain but loop. For the first time, there is a frontier, a possible circulation, a potential inside. The triangle is the weft, that which holds together (V346 p. 178). The tetrahedron is closure become volume. For the first time, an inside really exists. The corpus names these levels  $\Omega^0$ ,  $\Omega^1$ ,  $\Omega^2$ ,  $\Omega^3$ , and the order of their appearance is not a convention: each level requires the previous one.

The narrative uses the word “closure” here in the intuitive sense. Book II will strictly distinguish the triangular boundary, the filled 2-simplex and the dynamical rule capable of passing from the one to the other: these three objects are not identical.

## **1.6 — The birth of matter**

Why matter? Because matter is the only means, for the impetus [élan], of having an inside. Before it, everything is surface, everything is link without interior. The tetrahedron fabricates the first inside, like an egg (V346 p. 1167). And in this inside, everything changes. What is enclosed can be stored, transformed, lost. Time 3 switches on. Dissipation is born exactly here, at the formation of the tetrahedron, neither before (there was nothing to wear out) nor after (matter is constituted). Matter is therefore mortal whereas its carriers are not. It is one of the most singular ideas of the Blob, and it must remain visible: the carrier is eternal, what it constructs is not. Matter is also that which has a proper duration, a rhythm, a breathing. Each structure carries its intrinsic time, and nothing can be imposed on it from outside without deforming it. The corpus ended up making a rule of method of this: one measures the proper time of a system before interrogating it, never the reverse (V346 p. 1907).

The coincidence “first inside = first dissipation” is a founding intuition of the narrative. V347 does not transform it here into a theorem: its status is decided solely in Book II.

## **1.7 — The inherited opposition: matter and antimatter**

The two lineages issuing from the scission each construct tetrahedra. But they construct them with opposite polarity. On one side what we call matter; on the other its mirror orientation, antimatter. Antimatter is not a defective or rare matter. It is the exact twin. Everything that one of the universes constructs, the other constructs inverted, and the

sum of the two remains zero. The twin brings no new information: it is entirely determined by ours, like a reflection (V346 p. 1077, fact iii). In our universe, the dominant lineage is that of matter; in the twin, that of antimatter. But the narrative does not say that our universe contains only matter: it says that it leans. The two lineages coexist everywhere they have been transmitted, in coherent domains that the dynamics does not mix (V346 p. 1028–1029).

## **I.8 — The bath**

When the two lineages meet, they annihilate. A structure and its mirror undo each other. But annihilation is not sterile. In the author's intuition, it produces the bath: an undifferentiated reserve, without polarity, that feeds all subsequent construction. What comes out of the bath are the products of annihilation, what we shall later call photons; what remains after the bath is the residue, the excess of the dominant lineage that annihilation could not pair, because it removes the symmetric and never the asymmetry (V346 p. 1204). The bath is therefore at once the end of one generation and the beginning of the next. It is the place where the universe recycles itself.

## **I.9 — Chirality**

Here is the place where the narrative must yield before mathematics, and it is better to say so here than to hide it in Book II. The author first thought that the opposition of the two lineages would suffice to produce, in each universe, a net preference for one hand: that viability, by selecting what persists, would also select a chirality. The corpus showed the contrary, and it showed it against the initial desire: viability is blind to parity. A configuration and its mirror image have exactly the same viability. No selection, no viability-driven mitosis can create or erase a hand (V346 p. 1077, 1204).

The narrative survives this correction, but transformed. Orientation does exist, but it is not fabricated by selection: it is inherited from the genealogy, from the lineage born at the first scission, which viability never rewrites. And for this genealogical orientation to effectively couple to what is constructed, something must be posited, a coupling, whose origin the narrative does not supply. Book I says: the opposition exists from the genealogy onward. Book II will say: the viability functional does not transport the sign; a lineage variable and a coupling are necessary, and they are posited.

## **I.10 — Everything appears, not everything persists**

This is perhaps the deepest principle of the Blob, and the one that gave it its name. The impetus tries everything. It pushes, it overflows, it tries all possible configurations, continuously, without a plan. Most of these attempts lead nowhere: they come undone as soon as they are formed. But some hold. And what decides which ones hold is viability (V346 p. 1167). Viability is not a force. It does not create the attempts and does not constrain them. It is a filter of existence: what has the form of persistence is conserved and reproduced; what does not have it is undone, recycled, transformed. The universe is therefore not constructed because a final plan was inscribed at the beginning, but through three things together: variation, which proposes; viability, which retains; memory, which transmits what has been retained. And this selection has a price. Nothing structures itself for free. For a configuration to hold, order must be paid for in disorder

rejected into the surroundings. This is how the arrow of time 3 is born: not as a decree, but as the invoice of every structure that succeeds.

In the narrative, “everything appears” signifies the absence of a final plan and the richness of variation; it does not yet presume that the mathematical state graph is irreducible. Likewise, “to persist” is a cosmogonic concept for which Book II must supply an independent definition.

### **I.11 — Construction and continuous victory**

The universe is then constructed through a loop that does not stop: scission → construction → encounter → annihilation → bath → new construction, under the control of viability at each turn. At each generation, the matter born from the bath is more viable than that which preceded it, because what did not hold has been recycled and what did hold has been transmitted. This is what the author calls the continuous victory of matter: not a victory won once, but a victory replayed at each turn, through residue. There is a hierarchy in this loop. The tetrahedra assemble into larger structures which, in their turn, are subjected to the same sieve: test, keep, transmit. The same mechanism repeats at each level; only the brick and the binder change. And when this construction is looked at from afar, erasing the details, it must resemble itself: the large structures obey the same law as the small ones. This is what the corpus ended up calling G0, the renormalization gate, and it is the place where the program is working today.

### **I.12 — The resulting world**

Up to here, no word of the real has been necessary. The narrative has not needed Maxwell, Einstein, the Standard Model, Planck or  $\Lambda$ CDM. The reader must be able to stop here, telling themselves: I understand the world the author had in mind, before knowing whether it resembles ours.

The question of whether it resembles it is another question, which belongs to another book. It will be posed in a strict form: for each object constructed in Books I and II, what does it resemble in the observed world? And it will admit only four answers: identity, compatibility, analogy, failure. Nothing of what is found there will be able to come back and modify what has been said here.

### **I.13 — The why**

The corpus contains a chapter that the author wrote in the first person, and which says what drives the program: the impression of an impetus to mutate, to surpass itself, to occupy all available space; viability as a tipping point; matter as the shelter of the impetus; the living as the moment when viability ceases to be undergone and comes to be sought; consciousness as the organ through which the impetus turns back upon itself (V346 p. 1167–1169). V347 keeps this chapter in its place, in its register, that of openness and not of demonstration. It has no twin in Book II, and it will not have one. It is the only deliberately asymmetric pair of this book.

## **BOOK II — THE SAME STORY, IN MATHEMATICS**

Register: formal. II.0 is a dependency map and may announce objects defined further on. In the other chapters, every dependency must be explicit.

II.0 — The three formal strata and the two bridges

The corpus does not contain a single dynamics running seamlessly from  $\Omega^0$  to G0. It contains three historical strata that must be distinguished before any causal reading.

A\_gen — cosmogonic generator

Space of finite abstract simplicial complexes, lineage variable  $\sigma$ , mitosis operator  $\mu$ , variable number of vertices. This is the stratum that carries the genealogy and the growth narrative. Its current limit is exact:  $\mu$ , as defined in II.3, adds one leaf vertex and a single edge; it contains no closure rule.

A\_Gibbs — historical selection engine [ARCH]

At fixed N and  $|E|$ : geometric/Rips graph, clique (flag) complex, functional

$$S_{\text{sim}}(C) = \log(1 + n_{\text{tri}}(C)) - \beta D(C),$$

Gibbs measure and Metropolis edge exchanges. This engine can reorganize edges between existing vertices; it does not constitute, on its own, the genealogical mitosis of A\_gen.

B — canonical G0 sector

Finite simplicial triangulations of  $T^3$ , action

$$S(T) = \mu N_3 + \sum_e f(n_e),$$

and R-TRUE coarse-graining. Two subsectors are used:

- **B\_shell**:  $N_0$  fixed, Pachner  $2 \leftrightarrow 3$ ; conditional sector of the C2-R campaign.
- **B\_full**:  $N_0$  variable, Pachner  $1 \leftrightarrow 4$  and  $2 \leftrightarrow 3$ ; necessary for the inter-sector connections and for the first-passage kernels.

The old AUT-LIFT / full Pachner tension is no longer a fundamental no-go. V346 p. 1681 reports that an equivariant canonical recompaction makes simultaneously possible  $1 \leftrightarrow 4$  and a history-independent multiplicity; the 60,000-step bench tracks the quotient target at  $L1 \approx 10^{-3}$ . The status is operational, with a formal detailed-balance certificate still to be completed. This piece falls under RO-16/C4, not under a conceptual incompatibility between B\_shell and B\_full.

| Stratum        | Objects                                           | Dynamics                      | Role                 |
|----------------|---------------------------------------------------|-------------------------------|----------------------|
| A_gen          | abstract complexes, $\sigma$ , $\mu$ , variable N | partial genealogical growth   | origin/cascade       |
| A_Gibbs [ARCH] | Rips/flag graph at fixed N                        | edge exchange + Gibbs         | historical selection |
| B_shell        | $T^3$ triangulations, $N_0$ fixed                 | Pachner $2 \leftrightarrow 3$ | conditional C2-R     |

| Stratum | Objects                              | Dynamics                                              | Role                           |
|---------|--------------------------------------|-------------------------------------------------------|--------------------------------|
| B_full  | $T^3$ triangulations, $N_0$ variable | Pachner $1 \leftrightarrow 4$ , $2 \leftrightarrow 3$ | C4/first-passage/inter-sectors |

Two bridges are absent:  $A_{\text{gen}} \rightarrow A_{\text{Gibbs}}$  (RO-25) and  $A_{\text{Gibbs}} \rightarrow B$  (RO-08). As long as they are not constructed, the cosmogony and the canonical sector are a genealogy of models, not yet a single dynamics derived from end to end.

OPEN RESEARCH — RO-25: growth  $\rightarrow$  Gibbs bridge. Missing: a joint dynamics at variable  $N$  that combines mitosis/closure and relaxation in such a way that the conditional law at fixed  $N$  reproduces, or derives, the measure  $\pi_N$  of  $A_{\text{Gibbs}}$ . Falsifier: no local Markovian extension simultaneously possesses the desired genealogy, the announced conditional measure and a non-arbitrary closure rule. Proposed ledger: GROWTH-GIBBS-BRIDGE.

OPEN RESEARCH — RO-08:  $A_{\text{Gibbs}} \rightarrow B$  bridge. Missing: send the  $\Omega^3$  regime of the historical layer onto a closed triangulation of controlled topology, or prove that  $B$  is a typical/limit sector of  $A_{\text{Gibbs}}$ . Today  $T^3$  is imposed. Falsifier: typical  $\Omega^3$  configurations do not form closed 3-manifolds, or their topology is not controllable. References: V346 pp. 159–161, pp. 1243–1274, p. 1670. Ledger: LAYER-BRIDGE.

## II.1 — The initial object

Twin of I.1.

Definition 1.1 ( $\Omega^0$ ) [DEF].  $\Omega^0$  is the complex with one vertex and no simplex of higher dimension. In the category of non-empty pointed simplicial complexes, it is the one-point object equipped with the Dirac measure  $\delta_\star$  (V346 pp. 164–165).

Theorem 1.1 (initiality) [T]. In the pointed category under consideration, there exists a unique pointed morphism  $\Omega^0 \rightarrow K$  for every pointed object  $K$  (V346 p. 165).

Theorem 1.2 (triviality) [T].  $H_0(\Omega^0; \mathbb{Z}) = \mathbb{Z}$ ,  $H_n = 0$  for  $n > 0$  and  $\pi_n(\Omega^0) = 0$  (V346 p. 165).

Theorem 1.3 (absence of breakable structure) [T].  $\text{Aut}(\Omega^0)$  is trivial; no non-trivial internal involution is available (V346 p. 166).

Axiom A2 [A]. The framework does not derive the origin of  $\Omega$ ; it studies a dynamics once the configuration space has been posited (V346 p. 160).

Domain point [OPEN].  $D(C) = \text{std}(\text{deg})/E(\text{deg})$  is defined in V347 only when  $E(\text{deg}) > 0$ . At  $\Omega^0$ ,  $E(\text{deg}) = 0$ . Consequently, the old sentence “ $S_{\text{sim}}(\Omega^0) = 0$ ” is not a theorem of definition A3 without an explicit frontier convention. V347 does not add it silently; this convention is part of RO-02.

Correspondence with I.1. Uniqueness, the absence of internal orientation and topological triviality are carried. “Eternity” and “coldness” are narrative readings; no time variable,

nor  $S_{\text{sim}}$ , is yet defined without a frontier convention. Residue: structural CARRIED / WEAKENED for potentiality and coldness.

OPEN RESEARCH — RO-01: first movement. The framework does not derive what makes  $\Omega^0$  pass to a non-trivial configuration; A2 leaves it outside the framework.

## II.2 — The lineage variable

Twin of I.2.

Definition 2.1 (genealogy). Let  $\mathcal{G}$  be the directed acyclic graph of the vertices generated by mitosis (defined in II.3), an oriented edge  $v \rightarrow v'$  meaning that  $v'$  was born from  $v$ . A lineage is a function  $\sigma : V(\mathcal{G}) \rightarrow \{-1, +1\}$  such that  $\sigma(v') = \sigma(v)$  for every edge  $v \rightarrow v'$  (the child inherits from the parent).

Split axiom [A]. At the first division, the two immediate descendants receive opposite lineages:  $\sigma_1 = -\sigma_2$ . The choice of which of the two receives  $+1$  is a binary variable  $\sigma_0 \in \{-1, +1\}$ , posited, not derived (V346 p. 1077: “exactly one bit posited at the first division, everything else being derived”).

Proposition 2.2 (global neutrality) [T, trivial]. Under the split axiom and heredity, if the initial split is followed by the same number of mitoses in each lineage,  $\sum_v \sigma(v) = 0$ . More generally, the sum is determined by the cardinalities of the two subtrees alone.

Measurement 2.3 [M][ARCH]. On the cascade regime ( $N = 1500$ , 3 seeds), the fraction of vertices with  $\sigma = +1$  is  $0.498 / 0.494 / 0.512 \approx 0.50$ ; the fraction of interface edges (connecting two opposite lineages) is 5-6%. The edge-exchange dynamics does not mix the two lineages, which remain coherent in two domains at  $\sim 94\%$  (V346 p. 1028, §96.1).

Proposition 2.4 (immunity) [T, structural][ARCH].  $S_{\text{sim}}$  depends only on the current graph ( $n_{\text{tri}}$ , degree dispersion), never on  $\sigma$ . Any dynamics driven by  $S_{\text{sim}}$  leaves  $\sigma$  invariant: the lineage is neither created, nor erased, nor rewritten by selection (V346 p. 1028, 1077).

Correspondence with I.2. The formalism carries an inherited binary label and an opposite initial pair; it does not construct two complete universes.  $\sigma$  partitions vertices of one and the same configuration, while the twin spectral sector of II.7 remains [ARCH]. Global neutrality furthermore requires a mirror evolution of the two lineages, not heredity alone. Residue: WEAKENED; ABSENT are the dynamic double split and a globally twin state space.

OPEN RESEARCH — RO-03: derivation of the bit  $\sigma_0$ . Missing: a mechanism, internal to the framework, that fixes  $\sigma_0$  rather than drawing it. The corpus has tested seven native routes of global breaking, all washed out by  $S_{\text{sim}}$  (V346 p. 1028, cross-reference to Part XCV); none determines  $\sigma_0$ . Falsifier: if every internal determination of  $\sigma_0$  contradicts 1.3 (absence of breakable structure in  $\Omega^0$ ), the question is closed by theorem and  $\sigma_0$  remains an axiom. This is the most probable outcome, and it would be a result, not a failure.

Reference: V346 p. 1077 (CXVI), Core 1.0 §0.1.

OPEN RESEARCH — RO-04: two simultaneous simplices. Missing: the narrative has two opposite simplices born from a single act. The only formalized operator (mitosis  $\mu$ , II.3) generates one vertex at a time. No “double split” operation is defined in V346. Falsifier: show that every double-split operation is equivalent to two successive mitoses with opposite lineages (in which case the intuition is carried by 2.1 and the gap disappears), or that it requires a new object. Reference: V346 pp. 171–172 ( $\mu$ ), p. 1028 (split of the substrate).

OPEN RESEARCH — RO-05: the lineage in the canonical sector. Missing: the variable  $\sigma$  is defined only on layer A. The triangulations of  $T^3$  (layer B) carry no lineage, and the Pachner moves  $2 \leftrightarrow 3$  neither create nor destroy a vertex:  $\sigma$  has there neither genealogical support nor transport rule. The R-TRUE contraction (II.11) destroys a vertex: the inverse heredity rule (what becomes of  $\sigma$  under  $\pi : T \rightarrow U$ ?) is not defined. Falsifier: impossibility of defining  $\sigma$  compatible with the simplicial quotient. Reference: V346 p. 1670 (sector), p. 1914 (R-TRUE); depends on RO-08.

OPEN RESEARCH — RO-26: global twin state space. Missing: two configuration spaces  $\Omega_+$ ,  $\Omega_-$  (or a doubled space) connected by an involution  $J$ , with interlaced dynamics and a global neutrality rule; a simple label  $\sigma$  on a single graph is not enough. Falsifier: no global involution compatible with the dynamics and the coarse-graining exists. Proposed ledger: TWIN-STATE-SPACE.

## II.3 — Mitosis and conservation

Twin of I.3.

Definition 3.1 (simplicial mitosis) [DEF]. For  $C = (V, \Sigma) \in \Omega$  and  $v_0 \in V$ ,  $\mu_{\{v_0\}}(C) = (V \cup \{v'\}, \Sigma \cup \{\{v'\}, \{v_0, v'\}\})$  with  $v' \notin V$ , defined up to isomorphism (V346 p. 171).

Theorem 3.2 [T].  $\mu_{\{v_0\}}$  is well defined up to isomorphism, increases  $|V|$  and  $|E|$  by exactly 1, and does not raise the dimension beyond  $\max(\dim C, 1)$  (V346 p. 172, T2.1).

Theorem 3.3 (uniqueness of the first mitosis) [T]. Up to isomorphism, there exists a unique mitosis  $\Omega^0 \rightarrow \Omega^1$ ; its image is the 1-simplex  $\Delta^1$  (V346 p. 172, T2.2, consequence of 1.1).

Theorem 3.4 (combinatorial reversibility) [T]. The collapse of a degree-1 vertex is an inverse of  $\mu$  on  $\Omega^1$ :  $\mu^{-1} \circ \mu = \text{id}$ . Mitosis, as a combinatorial operation, can be undone (V346 pp. 172–173, T2.3).

Definition 3.5 (cardinal time  $t_1$ ) [DEF]. For  $C$  reached from  $\Omega^0$  by  $k$  mitoses,  $t_1(C) = k$ . On arborescences,  $t_1 = |V| - 1$  (V346 p. 172).

Proposition 3.6 (exact scope of vertex conservation) [T]. In  $A\_Gibbs$ , edge exchange preserves the set  $V$ ; in  $B\_shell$ , Pachner  $2 \leftrightarrow 3$  preserves  $N_0$ . In  $B\_full$ ,  $1 \leftrightarrow 4$  creates a vertex and  $4 \leftrightarrow 1$  destroys one.  $B\_full$  is used as a state graph/sampler and for first-passage; as long as  $Q\_phys$  is not constructed, V347 does not interpret  $4 \leftrightarrow 1$  as the physical death of a carrier. R-TRUE can also remove a vertex, but this is explicitly a change of scale. Thus the

“eternity of the carrier” is carried by A\_gen as posited genealogical memory and by the sectors at fixed  $\Omega_0$ , not as an invariant of all the formal dynamics of the corpus.

Correspondence with I.3. The “inscription” is carried by the genealogy and the counter  $t_1$  as long as no operation erases this provenance. But  $\mu$  is combinatorially reversible, B\_full contains  $4 \leftrightarrow 1$  and R-TRUE removes a vertex under coarse-graining: “eternity” is therefore not a universal invariant of all the operators of the corpus. It remains a narrative/sectoral property, not a general law. Residue: WEAKENED for eternity; CARRIED for local genealogy; ABSENT for nucleation.

OPEN RESEARCH — RO-02: nucleation  $\Omega^0 \rightarrow \Omega^1$  and frontier convention for D. Missing: (a) an explicit extension of D/S\_sim to  $\Omega^0$ , where the mean degree is zero; (b) a nucleation mechanism. The cascade from strictly empty  $\Omega^0$  is not spontaneous under the canonical engines: 0/40 seeds with the fixed-|E| engine and 0/40 with the grand-canonical engine, because creating an isolated edge does not increase S\_sim (no triangle) and D penalizes (V346 p. 174, caveat to M-2.1). The initial measurement “P = 1.00 on 8 seeds” relied on a non-empty germ. Falsifier: a preregistered nucleation protocol which, on  $\geq 30$  seeds, gives  $P(\Omega^1 | \Omega^0) \geq 0.9$  from the strict void. Reference: V346 p. 167 (H1.2), p. 174 (M-2.1).

## II.4 — The three orders and the two irreversibilities

Twin of I.4.

Definition 4.1 ( $t_0$ ) [DEF].  $t_0$  denotes the absence of non-trivial order on  $\Omega^0$ ; it is not a clock.

Definition 4.2 ( $t_1$ ) [DEF].  $t_1(C)$  is the number of mitoses along a given genealogy. It is an event counter, not a physical duration.

Definition 4.3 (sampler  $t_2$ ) [DEF]. For  $E(\deg) > 0$ ,  $D(C) = \text{std}(\deg)/E(\deg)$  and, along a sampler trajectory,  $t_2(n) = \sum_{k \leq n} D(C_k) \Delta t$  (V346 p. 161).

Identity 4.4 [T].  $dt_2/dt = D(C)$  is definitional. The historical  $R^2 = 0.999999$  is an implementation check, not a measured law.

Correction 4.5 [T]/[R]. For the regular 1-simplex  $\Delta^1$ ,  $D=0$ . But an irregular tree can have  $D>0$ . Since  $t_2$  is defined as a sum of D along the sampler, any repeated visit to states with  $D>0$  increments  $t_2$ . The old general statement “no dissipative arrow at  $\Omega^1$ ” therefore does not follow from the definition of D; it is exact only on regular subclasses or under a different notion of dissipation, still to be defined via Q\_phys. The universal version is retracted and absorbed by RO-23.

Requalification 4.6-a [T].  $\Omega^0$  contains no derivable internal temporal asymmetry (1.3).

Requalification 4.6-b [T/ARCH]. Certain regular pre-tetrahedral configurations used by the companion book have  $D=0$ ; this extends neither to all of  $\Omega^1$  nor to all of  $\Omega^2$ .

[R] 4.6-c — universal dissipative threshold at  $\Omega^3$ . The historical formulation “dissipation is born strictly and universally at  $\Omega^3$ ” is not proven by 4.5–4.6: configurations of dimension  $\leq 2$  can have  $D>0$  under the sampler definition. The only robust statement is conditional on the class of configurations considered. The general threshold becomes RO-23.

[R-ALG] 4.7-a — historical sign of the free energy. V346 defines  $H(\nu) = -\sum \nu \log \nu \geq 0$  and  $F[\nu] = -E\nu[S] + \tau H(\nu)$ , while asserting  $\pi_N \propto \exp(NS/\tau)$  and  $F[\nu] - F[\pi_N] = \tau \text{KL}(\nu || \pi_N)$ . These three expressions are incompatible. The correct sign in front of  $H$  is negative.

Theorem 4.7-b (KL Lyapunov) [T][ARCH]. For  $\pi_N \propto \exp(NS_{\text{sim}}/\tau)$ ,

$$\mathcal{L}[\nu] := \tau \text{KL}(\nu || \pi_N) = -N E\nu[S_{\text{sim}}] - \tau H(\nu) + \tau \log Z_N \geq 0.$$

For every kernel  $K$  leaving  $\pi_N$  invariant,  $\mathcal{L}[\nu K] \leq \mathcal{L}[\nu]$  by contraction of the relative divergence. This property is mathematical; it does not turn the sampler into a physical dynamics.

Measurement 4.8 [M][ARCH]. In the historical experiments, moves with  $\Delta S < 0$  concentrate more on sites carrying tetrahedra (reported factor  $\approx 3.3$ , four seeds,  $N=3000$ ). This measurement is not sufficient to establish a universal threshold.

Definition 4.9 (two historical regimes) [ARCH]. The corpus operationally distinguishes a pre-tetrahedral regime and a tetrahedral regime. This distinction is retained as a historical classification, not as a general theorem of dissipation. The value  $\tau_c \approx 0.40$  is valid only within a parametric slice; other  $\kappa$  give 0.963/0.470/0.274. V347 withdraws all universality from  $\tau_c$ .

Failure 4.10 (Q-TIME) [FAIL]. The stationary measure fixes neither the physical generator, nor its spectrum, nor a clock.  $Q_{\text{samp}}$  and  $Q_{\text{phys}}$  must be separated (V346 p. 1648).

Correspondence with I.4. The narrative of the three times translates formally into an absence of order, a genealogical counter and an accumulation tied to the sampler. The  $t_1/t_2$  separation is real; the identification of  $t_2$  with a physical time and the universal ignition at  $\Omega^3$  are not. Residue: CORRECTED.

RO-06 —  $Q_{\text{samp}} \neq Q_{\text{phys}}$ . Construct a kinetic generator independent of the sampler. Falsifier: impossibility of simultaneously satisfying locality, the desired stationarity and distinct kinetics.

RO-07 —  $t_1/t_2$  articulation and localization. Determine the link between genealogical counter and dissipation; reproduce or refute the historical localization correlation.

RO-23 — general dissipative threshold. Missing: characterize, over the whole admissible domain of dimension  $\leq 2$  and then 3, the relevant dissipative quantity; distinguish the sampler's  $D$  from a physical entropy production. Falsifier of the strong version of the narrative: an admissible configuration of dimension  $\leq 2$  exhibiting positive physical dissipation under  $Q_{\text{phys}}$ . Dependencies: RO-06, RO-27. Proposed ledger: DISSIPATION-THRESHOLD.

## II.5 — The four levels: object, boundary, filling

Twin of I.5.

Definition 5.1 [DEF].  $\Delta^0, \Delta^1, \Delta^2, \Delta^3$  are the abstract simplices of dimensions 0, 1, 2, 3. The historical symbols  $\Omega^0 \dots \Omega^3$  are retained in V347 as classification strata by maximal

dimension, and not as already-proven orbits of  $\mu$ . Their mathematical existence says nothing, by itself, about the dynamics that generates them.

Theorem 5.2-a ( $\mu$ -only NO-GO) [T]. Starting from  $\Delta^0$  and applying only the mitosis  $\mu$  of 3.1, each step adds a new vertex connected by a single edge to an existing vertex and adds no simplex of dimension  $\geq 2$ . By induction, the 1-skeleton remains a tree. Hence  $\mu$  alone can produce neither a cycle, nor  $\Delta^2$ , nor  $\Delta^3$ .

[R] 5.2-b — “ $\mu \circ \mu \circ \mu : \Omega^0 \rightarrow \Omega^2$ ” without an additional operation. V346 p. 177 writes this composition “with closure constraint”. This constraint is precisely a second mechanism: it is not contained in the definition of  $\mu$ . V347 therefore withdraws this expression as a derivation by  $\mu$  alone.

Topological correction 5.3 [T]. The boundary  $\partial\Delta^2$  is homeomorphic to  $S^1$  and carries  $\pi_1 \cong \mathbb{Z}$ . The filled 2-simplex  $\Delta^2$  is a disk, hence  $\pi_1(\Delta^2)=0$  and  $H_1(\Delta^2)=0$ . The old language “the triangle  $\Delta^2$  is the first non-trivial  $\pi_1$ ” confuses boundary and filling.

Flag corollary 5.3-bis [T][ARCH]. In a clique/Rips complex, every 3-clique  $K_3$  is filled by a 2-simplex. A triangular cycle therefore does not create  $H_1$ ; the first possible unfilled cycle in a flag complex is an induced cycle of at least four vertices. This takes nothing away from the fact that  $\Delta^2$  is the first 2-simplex; it only withdraws the old topological interpretation.

Theorem 5.4 (first 3-simplex) [T].  $\Delta^3$  is the smallest simplex of dimension 3. In a non-degenerate geometric realization, it has a three-dimensional interior and a positive volume. In the purely abstract framework, the exact statement is only “first 3-simplex”.

Historical Theorem 5.3-bis ( $K_3=15/32$ ) [T][ARCH]. The coefficient  $15/32$  concerns the clustering of the historical 3D geometric substrate, not the topology of  $\Delta^2$ . It remains archived within its correct scope.

Identities 5.5 (layer B) [T]. For any closed triangulation of a 3-manifold:  $N_2=2N_3$ ,  $N_1=N_0+N_3$  and  $\sum_e n_e=6N_3$ .

Theorem 5.6 (I2-MIN) [T].  $I_2=\sum_v q_v^2/(16N_3^2) \geq 1/N_0$ , with equality if and only if the valences are uniform.

Correspondence with I.5. The hierarchy of simplicial dimensions is carried. On the other hand, the causality “mitosis closes and successively generates triangle then tetrahedron” is not carried by  $\mu$  as it is defined. And the first topological loop is not the filled 2-simplex. Residue: CORRECTED for the topology; ABSENT for the closure mechanism.

OPEN RESEARCH — RO-24: closure generator. Missing: an operation, already justified by the corpus or explicitly added in a later version, capable of connecting existing vertices and of specifying when the flag complex fills cliques; prove how it interlaces with  $\mu$  without introducing an ad hoc plan. Most economical option to test: interleave the already existing edge dynamics of  $A\_Gibbs$  between growth events, rather than inventing a third law. Falsifier: no local closure rule compatible with viability and provenance reproduces the announced cascade. Proposed ledger: CLOSURE-GENERATOR.

## II.6 — The $\Omega^3$ threshold and the canonical sector

Twin of I.6.

[R] 6.1-a — “ $S_{\text{sim}}$  is zero on  $\Omega^0$  and  $\Omega^1$  and becomes non-trivial at the triangle/tetrahedron”. This historical formulation is too strong. A3 contains two terms: the triangular reward and the dispersion penalty. On an irregular tree,  $n_{\text{tri}}=0$  but  $D>0$ , hence  $S_{\text{sim}}=-\beta D \neq 0$ . At  $\Omega^0$ ,  $D$  is not even defined without a frontier convention.

Proposition 6.1-b (separate thresholds) [T].

- on regular  $\Delta^1$ :  $n_{\text{tri}}=0$  and  $D=0$ , hence  $S_{\text{sim}}=0$ ;
- on an irregular tree: the triangular term remains zero but the penalty  $D$  can be non-zero;
- at the first filled triangle:  $\log(1+n_{\text{tri}})>0$  for the first time;
- at  $\Delta^3$ : a 3-simplex exists for the first time.

These four thresholds are distinct. V347 henceforth forbids merging them into a single “non-triviality threshold”.

Definition 6.2 ( $B_{\text{shell}}$ ) [DEF]. Finite simplicial triangulations of  $T^3$  at fixed  $N_0$ , Pachner  $2 \leftrightarrow 3$ . The topology  $T^3$  is imposed, not derived.

Definition 6.3 (production action) [DEF].  $S(T)=\mu N_3 + \sum_e f(n_e)$ ,  $f(n)=-\alpha \log(1+n)+b(n-d\star)^2$ ,  $d\star=2\pi/\arccos(1/3)$ . The point  $\alpha=0.4$ ;  $b=0.05$ ;  $\mu=3.0$  is a frozen production choice, not a fundamental constant.

Definition 6.4 (germ) [DEF/CERT]. Periodic Freudenthal-Kuhn germ; manifold checks,  $\chi=0$ , valences and  $I_2$  recorded in V346.

Theorem 6.5 (HODGE-RANK) [T]. For any finite triangulation of  $T^3$ ,  $\dim \ker \Delta_1 = b_1(T^3) = 3$ .

Measurement 6.6 [M][ARCH]. The historical Hodge rigidity is higher on the edges engaged in tetrahedra under  $S_{\text{sim}}$  than in two controls; this result is not transported to B.

Correspondence with I.6. The “first inside” is carried by the first 3-simplex and, in B, by the tetrahedra of a 3-manifold. The sentence “dissipation is born exactly here” remains conditional and refers to RO-23; the identification of the 3-simplex with physical matter is deferred to the Book of Confrontation. Residue: CARRIED for simplicial volume; WEAKENED/CORRECTED for dissipation and physical identification.

RO-19 — imposed topology.  $T^3$  remains a sector of study, not an output of the cascade.

RO-17 — dissipation on  $T^3$ . Redo the historical dissipation/rigidity diagnostics under the canonical action, with controls and intervals.

## II.7 — The involution and the two sectors

**\*\*Definition 7.1 (lineage involution).**  $\mathcal{C} : \sigma \mapsto -\sigma$  on  $\mathcal{G}$ . By 2.1,  $\mathcal{C}$  exchanges the two subtrees born from the split.

**Theorem 7.2 (spectral symmetry) [T][ARCH].** In the Dirac–Kähler model on the layer-A graph, the operator  $D$  and the grading  $\Gamma$  satisfy  $\{D, \Gamma\} = 0$  (structural zero, 3 seeds), whence a mirror spectrum: to every mode  $+\lambda$  there corresponds exactly one mode  $-\lambda$ ,  $\max|\lambda + (-\lambda)| \approx 2 \cdot 10^{-14}$  (V346 p. 1077, fact ii).

**Theorem 7.3 (forced twin sector) [T][ARCH].** With  $H(\sigma) = D + g \cdot \sigma \cdot \Gamma$ , one has  $\Gamma \cdot H(\sigma) \cdot \Gamma = -H(-\sigma)$  to machine zero. The  $-\sigma$  sector is therefore entirely determined by the  $\sigma$  sector: it brings no independent information, and the sum of the charges of the two sectors is zero by identity, not by adjustment (V346 p. 120, 1077 fact iii).

**Measurement 7.4 (opposite domains) [M][ARCH].** With  $g = 1.5$ ,  $N = 350$ , 3 seeds: net charge  $-0.276$  in the domain  $\sigma = +1$ ,  $+0.271$  in the domain  $\sigma = -1$ ,  $+0.003$  globally. Decisive control: a mixed lineage (same signs, dispersed) gives  $+0.315$  and  $+0.675$ , of the same sign, without twinning. The coherence of the  $\sigma$  domains is necessary; a local sign is not sufficient (V346 p. 1029, §96.2). Posited variable 7.5 [A][ARCH]. The coupling  $g \cdot \sigma \cdot \Gamma$  is posited:  $g$  is a parameter,  $\sigma \cdot \Gamma$  the term odd under  $\mathcal{C}$ . Theorem 9.3 (below) requires that such a coupling be supplied from outside  $S_{\text{sim}}$ ; it does not derive it. The lineage alone is inert: its correlations with curvature, tetrahedron density and orientation are  $\approx 0$  on all seeds (V346 pp. 1028–1029, §96.1 caveat, §96.3).

Correspondence with I.7. The “inherited opposition”, the “exact twin that brings no information”, the “global neutrality, local asymmetry” are carried (7.1–7.4) — but only in an archived model, with a posited coupling. None of this exists on layer B. Residue: WEAKENED [ARCH].

OPEN RESEARCH — RO-09: lineage–structure coupling in a non-archived formalism. Missing: (a) an object playing the role of  $\Gamma$  (grading odd under  $\mathcal{C}$ ) defined on the layer-B triangulations; (b) a derivation, or at least a selection principle, for  $g$ . The corpus has located the coupling “in the right place” (in the odd term, not in the carrier) without eliminating it (V346 p. 1029, §96.3). Falsifier: if every term odd under  $\mathcal{C}$  is necessarily zero on layer B (for example because the Pachner moves commute with every admissible involution), local asymmetry does not exist in the canonical sector. Reference: V346 pp. 1028–1029, 1077; depends on RO-05.

The archival  $H(\sigma)$  structure formalizes two interlaced spectral sectors; it is not sufficient to construct two complete dynamical universes. This gap is carried by RO-26.

## II.8 — The reduction

8. This chapter is almost empty, and this nakedness is constitutive: it marks the exact place where the narrative goes beyond the corpus. What V346 contains. A narrative-measured mechanism, on layer A, in which: an integral memory  $B_{\nu} = \int m$ , carried by a sector exempt from the second irreversibility (defined on the  $t_0$  frontier, “ $d\tau = 0$ ”), biases the cascade; the symmetric pairing of configurations of opposite lineages removes the pairs, increments a counter  $B_{\gamma}$  and preserves the net excess; the analogue of the excess/bath ratio is  $\eta = (\text{cumulative net excess})/B_{\gamma}$  (V346 p. 1204,

CResidue: WEAKENED (weighting instead of filter; local exploration instead of exhaustiveness), on finite windows.

OPEN RESEARCH — RO-20: define “persist”. Missing: a definition of persistence independent of  $\pi_N(C) > 0$ , which makes axiom A0 non-empty: admissible set, zero-temperature limit, or survival criterion over a horizon. Book I depends on it entirely (“not everything persists”); Book II possesses only the weighted version of it. Falsifier: if every binary definition of persistence is equivalent to an arbitrary threshold on  $\pi_N$ , the filter does not exist and the narrative must be rewritten in terms of frequencies. Reference: V346 pp. 159–161 (A0, A4, A5), p. 165 (A0 posited as “generating principle”).

OPEN RESEARCH — RO-21: global accessibility. Missing: a proof of irreducibility of the state graph for each sector used (A; B\_shell; B\_full), and a statement of what is not accessible from the germ. The bipartition by volume parity (every Pachner move changes  $N_3$  by an odd number, V346 p. 1635) is already a known obstruction. Falsifier: classes of states inaccessible from the germ under  $2 \leftrightarrow 3$  alone. Reference: V346 pp. 1634–1635, 1654, 1661. \*\*OPEN RESEARCH — RO-12XIII.1). An out-of-equilibrium window [M] (the excess is born only during mitosis, ratio  $12.9\times$  versus equilibrium) and a sign discriminator [M] (only a coupling  $\kappa > 0$  locks in a stable excess;  $\kappa \leq 0$  remains at noise level) were measured on 4 to 6 seeds (V346 p. 53, V241). What V346 does not contain. No operation  $(C_+, C_-) \mapsto B$  defined on  $\Omega$ , no invariant conserved by this operation, no proof that B has zero lineage, no statement of what “feeds” means formally. The ratio  $\eta$  is not pinned to a value, and any cosmological value historically associated with it is in [R]/[FAIL] (Core 1.0 §4: cosmology excluded from the Core; Ledger COSMO-OLD).

Hypothesis 8.1 [H]. There exists a reduction operation  $\mathcal{R} : (C_+, C_-) \mapsto B$  on pairs of configurations of opposite lineages, such that (i)  $\sigma(B) = 0$  in the sense of a neutral lineage to be defined, (ii) an additive quantity (to be defined) is conserved by  $\mathcal{R}$ , (iii)  $\mathcal{R}$  acts only in regime 2 (4.9). Falsifier: impossibility of defining  $\mathcal{R}$  compatible with 3.6 (conservation of the number of vertices under the dynamics) — which would force placing  $\mathcal{R}$  in the coarse-graining (II.11) and not in the dynamics.

Correspondence with I.8. Residue: ABSENT. The intuition of the bath is one of the richest of the narrative and one of the least formalized of the corpus.

OPEN RESEARCH — RO-10: the reduction operation. Missing: the definition of  $\mathcal{R}$  (8.1) and of its invariant. Prior question: does  $\mathcal{R}$  belong to the dynamics (it then destroys vertices, contrary to 3.6) or to the coarse-graining (it is then an R-TRUE contraction marked by the lineage, which links RO-10 to RO-05 and II.11)? This second option is the most economical and the most consistent with the narrative (“the bath is the place where the universe recycles itself” = change of scale). Reference: V346 pp. 53, 1204–1205; Core 1.0 §4. Ledger: new entry to be created (BATH-OP, OPEN, [H]).

OPEN RESEARCH — RO-11: the ratio  $\eta$ . Missing: any value of  $\eta$  derived from the Core. The old numerical coincidences are retracted and cannot be rehabilitated without an independent derivation from the core (Ledger COSMO-OLD, Falsifier: “Independent derivation from Core only”). Status: forbidden to state a value before RO-10.

## II.9 — Parity blindness

Definition 9.1 (mirror). For an embedded configuration  $C$  (layer A),  $\text{mirror}(C)$  is the image of  $C$  under a reflection  $x \mapsto -x$  of the embedding; the net orientation is  $O_{\text{net}} = \sum_{\{\text{tetrahedra}\}} \text{sign}(\det)$ .

Proposition 9.2 (invariance of  $S_{\text{sim}}$ ) [T, structural].  $S_{\text{sim}}$  depends only on the abstract graph ( $n_{\text{tri}}$ , degree distribution), never on the coordinates. Hence  $S_{\text{sim}}(\text{mirror}(C)) = S_{\text{sim}}(C)$  identically. Verification: deviation of exactly 0.0 on 3 seeds, whereas the reflection inverts 100% of the orientations ( $+0.023 \rightarrow -0.023$ ) (V346 p. 1077, fact i; p. 50, V245: the reflection leaves the graph unchanged but inverts all chiralities).

Theorem 9.3 (nullity of the net orientation on average) [T]. On a uniform i.i.d. substrate and under any dynamics whose measure is reflection-invariant,  $\mathbb{E}[O_{\text{net}}] = 0$ : the reflection reverses each determinant and leaves the measure unchanged, whence  $\mathbb{E}[O_{\text{net}}] = -\mathbb{E}[O_{\text{net}}]$ . Complementary measurement [M]:  $\text{Var}[O_{\text{net}}] = \Theta(N)$ , hence  $|O_{\text{net}}|/N \sim N^{-1/2} \rightarrow 0$  (V346 p. 1204, Part CXXXI “law of the +tiny”).

Corollary 9.4 (no handedness preference by selection) [T]. No dynamics driven by  $S_{\text{sim}}$  can create, amplify or erase a net orientation. The left/right direction is flat in  $S_{\text{sim}}$ : it diffuses as a random walk around zero (V346 p. 1077). Complement 9.5 (attraction towards achirality) [M][ARCH].  $S_{\text{sim}}$  is not merely invariant: it is attractive towards symmetry. Seven native breaking routes lodged in the emerged structure (birth bias, spatial twin, mitosis screw, global substrate, bridge) were washed out by the dynamics (V346 p. 1028, cross-reference to Part XCV).

Measurement 9.6 (correlation without a net hand) [M][ARCH]. What the dynamics produces is an orientation correlation: two tetrahedra adjacent through a face more often have the same orientation than the opposite,  $P = 0.516$ , with net pseudoscalar  $\approx 0$ , flat in  $N$ , non-percolating domains (V346 p. 50, V245, rectification of CXXXVI.4). Measured mechanism: same-orientation pairs realize more triangles in their five-vertex configuration (9.29 versus 9.18), which favours them under  $S_{\text{sim}}$ ; geometric compactness is anti-correlated ( $-4.3\sigma$ ), hence the effect is carried by triangle counting (V346 p. 974, §78.2). Caveat on significance. The corpus reports  $22.6\sigma$ , growing with  $N$  ( $18.7\sigma$  at  $N = 3000$ ;  $24.3\sigma$  at  $N = 6000$ ), which is consistent with a small deviation measured over a very large number of pairs; but the pairs of a single configuration are not independent, and the number of  $\sigma$  is therefore overestimated. V347 retains the sign and the order of magnitude of  $P$ , not the  $\sigma$  count. The sign subsequently received an exact argument at first order in the intensity of viability (V346 p. 59).

Correspondence with I.9. The initial narrative (“selection chooses a hand”) is refuted by 9.2–9.4. The corrected narrative (“orientation is inherited from the lineage, not manufactured by selection, and a coupling must be posited”) is exactly what 9.2–9.4 + 2.4 + 7.5 say. The chain is coherent: (1)  $S_{\text{sim}}$  is blind; (2) hence polarity must be carried by a variable that  $S_{\text{sim}}$  does not rewrite ( $\sigma$ , 2.4); (3) hence its coupling to structure must be supplied from outside  $S_{\text{sim}}$  (7.5). Residue: CORRECTED, and it is the best-documented residue of the corpus.

## II.10 — The measure, viability and the kernel

Twin of I.10.

Historical A0 [A][R]. V346 wrote  $\pi(C)>0 \Rightarrow C$  viable. Under a finite-energy Gibbs,  $\pi(C)>0$  for every admissible configuration: this support defines no binary filter.

A0' (viability as weighting) [A].  $\pi_N$  assigns a relative stationary weight to the admissible configurations. Under ergodicity, these weights govern the long-term empirical residence frequencies. "Persist" is not defined by  $\pi_N(C)>0$ ; it requires a distinct criterion (RO-20).

A3 [A][ARCH].  $S_{\text{sim}}(C) = \log(1 + n_{\text{tri}}(C)) - \beta D(C)$ ,  $\beta = 0.30$ .

A4 [A][ARCH].  $\pi_N(C) = Z_N^{-1} \exp(NS_{\text{sim}}(C)/\tau)$ ,  $\tau = 0.50$  in the historical sector.

A5 [A][ARCH]. Metropolis edge exchange at fixed  $|E|$ , with the planned rejection of out-of-domain states.

Theorem 10.1 [T][ARCH]. The historical DLR/Dobrushin results remain archived within their domain; V347 does not automatically transport them to B.

B\_shell kernel. On  $T^3$  at fixed  $N_0$ : Pachner  $2 \leftrightarrow 3$ , Barker or Metropolis-Hastings, target  $\propto e^{-S(T)}$ .

Rule 10.2 (breathing) [METH]+[M]. For an observable of a stationary chain whose autocorrelation is readable, estimate  $\tau_{\text{int}}$  before fixing cadence and budget. At  $L=5$ , the corpus reports  $\hat{R}(N_4) = 1.01272$  with imposed cadence versus 1.00300 with adapted cadence, at budgets about 1.6% apart. This is a measurement of the sensitivity of the diagnostic to the protocol, not a universal theorem about "every system".

Certificate 10.3 (C2-R) [M][OPEN]. The L4/L5 windows and the inter-engine disagreement remain recorded as in V347.0. The contractual status is OPEN; no finite window counts as a thermodynamic limit.

Rule 10.4 (five checks) [METH]. Before any statistical reading: (0) algebraic identities; (0-bis) identity of the configuration; (1) sample count; (2)  $\tau$  readability condition; (3) stability under lengthening. The name "four checks" in V347.0 was a simple miscount.

Correspondence with I.10.

- "variation without a plan": CARRIED locally by the neighbour proposals;
- "everything appears": OPEN, because it requires accessibility/irreducibility (RO-21);
- "not everything persists": ABSENT as a binary criterion, the Gibbs only weights (RO-20);
- "memory": carried by the genealogy only in  $A_{\text{gen}}/A_{\text{shell}}$  under conditions;
- "the price of order is rejected into the surroundings": not proven by the decrease of KL alone; it requires an entropy production with system/environment (RO-27).

Global residue: WEAKENED.

RO-12 — confirmatory C2-R. Independent replication and size extension according to the frozen protocol.

RO-18 — finite window  $\rightarrow$  limit. No L4/L5 certificate is a thermodynamic limit.

RO-21 — global accessibility. Prove the irreducibility/connectivity of the relevant state graph or characterize its components exactly.

RO-20 — define “persist”. Define a persistence observable independent of the mere support of  $\pi$ : residence time, survival over a horizon, basin, or another preregistered criterion.

RO-27 — local entropic bookkeeping. Missing: a system/environment split and a non-negative entropy production for a dynamics  $Q_{\text{phys}}$ , making it possible to give a formal meaning to “order is paid for in rejected disorder”. The decrease of KL of a sampler is not sufficient. Dependency: RO-06. Falsifier: no local notion of entropy production compatible with the candidate dynamics. Proposed ledger: ENTROPY-BOOKKEEPING.

## II.11 — The change of scale

This chapter formalizes the “construction” and the “continued victory” not as an evolution but as a change of scale: what, in the narrative, is the passage from one generation of structures to the next is here the passage from a triangulation  $T$  to its contraction  $U$ , and the question is what survives this passage.

Definition 11.1 (R-TRUE). A simplicial contraction of an edge  $uv$  is legal if and only if  $\text{lk}(u) \cap \text{lk}(v) = \text{lk}(uv)$  as full simplicial complexes. On a closed 3-manifold, one controls the link defects  $E_0$  (vertices),  $E_1$  (edges),  $E_2$  (triangles); legality  $\iff E_0 = E_1 = E_2 = 0$  (V346 p. 1914; Core 1.0 §1.4). Retraction 11.2 [R]. The condition based on equality of the vertex sets of the links alone tests only  $E_0 = 0$  and does not certify contractibility. It overestimated the number of legal edges by 72% ( $L = 4$ ) and 81% ( $L = 5$ );  $E_1$  is the dominant filter (V346 p. 1914; Ledger VERTEX-LINK-CONDITION, superseded).

Theorem 11.3 (LINK-QUOTIENT-DEFECT) [T].  $\pi$  is injective on non-degenerate tetrahedra if and only if  $E_2 = 0$ , and  $N_3(U) = N_3(T) - d - E_2$  (V346 p. 1914).

Certificate 11.4 (legal set and f-vector) [CERT]. The complete filter reduces 334 candidates to 101 legal edges at  $L = 4$ , and 596 to 211 at  $L = 5$ . Independent check  $(\Delta N_0, \Delta N_1, \Delta N_2, \Delta N_3) = (-1, -(d+1), -2d, -d)$ : 101/101 and 211/211 (V346 p. 1914; Ledger RTRUE-LINK, RTRUE-FVECTOR, T/CERT).

Theorem 11.5 (exact contraction action) [T]. With  $C = V(\text{lk}(uv))$ ,  $E_C = E(\text{lk}(uv))$ :  $\Delta S_{\text{ctr}} = -\mu d - f(d) + \sum_{\{x \in C\}} [f(n_{ux} + n_{vx} - 4) - f(n_{ux}) - f(n_{vx})] + \sum_{\{xy \in E_C\}} [f(n_{xy} - 1) - f(n_{xy})]$ . The fusion term is  $-4$ , not  $-2$ : the triangle  $uvx$  belongs to two tetrahedra, counted four times in  $n_{ux} + n_{vx}$ . Checks  $\Delta N_1 = -(d+1)$  and  $\Delta \Sigma n_e = -6d$ : 101/101 and 211/211 (V346 p. 1915; Ledger RTRUE-ACTION, supersedes the old formula with  $-2$ ).

Measurement 11.6 (sign) [M][CERT window].  $\Delta S_{\text{ctr}} < 0$  on 101/101 and 211/211: every legal contraction lowers the action on the certified windows. The observed brake is topological, not energetic. A finite-window result, never a thermodynamic theorem (V346 p. 1915; Ledger RTRUE-SIGN, Next\_Action “Re-test L6/L7”).

Measurement 11.7 (channel  $d = 3 + 4$ ) [M]. Fraction of events of degree  $d \in \{3, 4\}$ : by count 0.9406 ( $L4$ ) / 0.9289 ( $L5$ ); by rate 0.998499 / 0.998089; by dissipation ( $\kappa = 1$ )

0.998432 / 0.998005. The surviving degrees  $\geq 5$  have inverse rates from  $10^{-5}$  to  $10^{-8}$ ; they are deprioritized, not asymptotically excluded (V346 p. 1915; Ledger MARKED34-RATE). Distinctions 11.8 [R of earlier confusions]. Activity  $\neq$  rate:  $u_e = z \cdot \rho_e \cdot e^{\{-\Delta S\}}$  (polymer activity, forward) versus  $r_e = e^{\{\Delta S_{ctr}\}}$  (rate of the inverse generator); with  $\Delta S < 0$ , one is  $\in [1.7 \cdot 10^2, 9.4 \cdot 10^6]$  and the other  $\in [7.5 \cdot 10^{-8}, 5.7 \cdot 10^{-3}]$ . Carrier  $\neq$  halo:  $|B_e| = d$  versus  $|H_e|$  7.5 to 7.9 times larger. “Contained in the carrier”  $\neq$  interior\_complete: the correct predicate is  $F_e = \pi(C_e) \subset X$  on a coarse domain closed under faces (V346 p. 1915; Core 1.0 §4).

Theorem 11.9 (SIMPLICIAL-PULLBACK GRADING) [T/CERT]. For  $X \subset U$  tetrahedral, closed under faces, and  $X' = \pi^{-1}(X)$ ,  $\Delta m = |X'_3| - |X_3| = \#\{\tau \in C_e : \pi(\tau) \in X\}$ , hence  $0 \leq \Delta m \leq d$ , with  $\Delta m = d \iff F_e \subset X$ . Four checks on 19,990 pairs (6,003 at L4, 13,987 at L5), zero violations (V346 p. 1916; Ledger PULLBACK-GRADING, supersedes three invalid transports).

Theorem 11.10 (sign of the crossings) [T]. For  $\kappa > 0$  and  $r_e = e^{\{\Delta S_{ctr}\}} > 0$ :  $c_e = r_e(1 - e^{\{-\kappa \Delta m\}}) \geq 0$  for every legal R-TRUE event transported canonically, including the boundary\_crossing ones. Hence  $B_X^{\text{bad}} = 0$  and  $O_{\{X, RTRUE\}}^{\text{bad}} = 0$  by theorem, and  $L_X(\kappa) \geq K_X + D_{34, X}(\kappa) \geq D_{34, X}(\kappa)$  (V346 p. 1916; Ledger CROSSING-SIGN).

Certificate 11.11 (local MARKED margin, finite) [CERT/M]. Six configurations (three seeds per size):

| L | seed | $\omega(R=2)$ | $\omega(R=3)$ | $\omega(R=4)$ |
|---|------|---------------|---------------|---------------|
| 4 | 1    | 0.1411        | 0.2520        | 0.5046        |
| 4 | 2    | 0.0458        | 0.1642        | 0.3334        |
| 4 | 3    | 0             | 0.0398        | 0.0398        |
| 5 | 1    | 0.0630        | 0.2633        | 0.6634        |
| 5 | 2    | 0.3673        | 0.5551        | 0.6186        |
| 5 | 3    | 0.0570        | 0.0570        | 0.6372        |

Positive sign on 6/6 at  $R = 3$  and  $R = 4$ ;  $\omega(R = 2) = 0$  on one configuration ( $R = 2$  is not universal); amplitude not reproducible (factor 6.3 at L4, 9.7 at L5); minimum 0.0398. The preregistered threshold  $\omega^* = 0.20$  was withdrawn before being posited (the +4.5% stability between L4 and L5 on seed 1 was a coincidence). By D126,  $\omega_R$  provides  $\|e^{\{tA_K\}}\|_w \leq e^{\{-\omega t\}}$  for the raw normalized marked operator, with the worst of the six (V346 p. 1916; Ledger FINITE-MARKED-GAP).

Theorem 11.12 (pointwise-uniformity no-go) [T].  $\inf_{\{T \in \Omega(T^3)\}} \omega(T) > 0$  is impossible. Proof:  $T^3$  admits finite triangulations, hence a triangulation with minimal  $N_0$ ; if it possessed a legal R-TRUE edge, its contraction would preserve  $T^3$  with  $N_0 - 1$  vertices, contradicting minimality; hence its legal set is empty and  $\omega = 0$ . Consequence: the thermodynamic theorem sought necessarily concerns a typical sector of the Gibbs measure, never every combinatorially possible triangulation (V346 p. 1917; Ledger POINTWISE-NOGO).

Target 11.13 (GEN-4, reformulated) [OPEN]. There exist  $R^\star$ ,  $\omega^\star > 0$  and typical sets  $\mathcal{G}_L$  with  $\nu_L(\mathcal{G}_L) \rightarrow 1$  such that  $\inf_{\{T \in \mathcal{G}_L\}} \inf_X D_{34, R^\star}(X; T)/s(T) \geq \omega^\star$ , with  $\|E_L\|_w \rightarrow$

0 on the complement, giving  $\|e^{\{tA_L\}}\| \leq e^{\{-(\omega^\star - \|E_L\|)t\}}$ . No value of  $\omega^\star$  is fixed by the data (V346 p. 1917; Core 1.0 §3.1; Ledger GEN4-TYPICAL).

Closed methods 11.14 [FAIL]/[R]. D102 (absolute weighted activity in the same norm):  $\Lambda \approx 1.16 \cdot 10^{13}$  at  $\kappa = 3$  after purging the channel, threshold 32.72, dead method, do not spend any budget on it. Concordance  $\Gamma_{\min}/s = 0.079 \longleftrightarrow 0.083$ : withdrawn (produced by the contaminated channel).  $\eta_{\text{all}}$  catastrophe: withdrawn. Three transports rejected (V346 p. 1917; Ledger D102, FAIL).

Contract G0, status 11.15 [OPEN]. C2-R (10.3); C3 multi-scale campaign not closed; C4 exact coarse-graining with  $\|D_m E_m - I\|$  not uniformly certified; C5 reduced fixed point (old basis disqualified,  $r_{\text{head}} \times 10^6$ ); C6 tail/locality (raw MARKED kernel finite and positive, uniformity and MARKED  $\longleftrightarrow$  CONNECTED open); C7 universality (several regulators exist, convergence not proven) (V346 p. 1918; Core 1.0 §3).

Correspondence with I.11. “The same sieve replayed at every level” is the R-TRUE contraction; “what has held is transmitted” is the pull-back 11.9 and the sign 11.10 (nothing that is transported canonically contributes negatively); “the continued victory” is, at best, the positive local margin 11.11 (what survives the change of scale contracts the tail). But 11.12 forbids the pointwise version of the narrative (“at every level, always”): the right formulation is “on the typical sector, with probability  $\longrightarrow 1$ ”. And 11.13 is open. Residue: CARRIED for transmission under change of scale; ABSENT for generational improvement. R-TRUE formalizes the survival of information under coarse-graining; it does not formalize “each generation is more viable than the previous one”. A contraction

RG is not a generation: the pull-back and the MARKED margin show that a transported structure has a favourable local drift, not a monotone improvement from generation to generation. The first piece of G0 falls without an asterisk, but it is not G0, and it is not the continuous victory.

OPEN RESEARCH — RO-22: generational improvement. Missing: a “generation” object (distinct from the scale) and a viability quantity comparable from one generation to the next, with a statement of monotonicity or non-monotonicity. Depends on RO-10 (the bath is the passage between generations) and on RO-20 (defining “persist”). Falsifier: a dynamics in which the mean viability of the structures issuing from the bath decreases. Reference: V346 p. 1167 (narrative), p. 1916 (local drift).

OPEN RESEARCH — RO-13: typical-set uniformity of the local margin. Missing: measure, then bound from above, the distribution of  $\omega$  at fixed R on the Gibbs sector as L grows, with norm control of the exceptional set. The dispersion by a factor of 10 shows its difficulty. Falsifier F-G0-2: no fixed  $R^\star$  gives a positive typical-set margin as L increases, or the exceptional complement does not vanish in norm. Reference: V346 pp. 1916–1917; Core 1.0 §3.1–3.2, §11 (step 4). Ledger: GEN4-TYPICAL (OPEN, “Primary theorem target”).

OPEN RESEARCH — RO-14: MARKED-SUPPORT  $\longleftrightarrow$  CONNECTED-POLYMER. Missing: an intertwiner (or a cumulant transform) establishing, in a declared Banach norm, that the connected polymer tail inherits the drift of the marked support. Falsifier F-G0-3: the

passage amplifies the norm to the point of cancelling the contraction. Reference: V346 p. 1917 (point B); Core 1.0 §3.2, §11 (step 5). Ledger: MARKED-CONNECTED (OPEN).

OPEN RESEARCH — RO-15: cutoff wall and projective limit. Missing: local convergence of the semigroup and stability under removal of the regulator, without reintroducing non-local modes. Falsifier F-G0-4: the limit does not converge or depends on the regulator. Reference: V346 p. 1918 (point C); Core 1.0 §3.2, §11 (step 6). Ledger: CUTOFF-PROJ (OPEN).

OPEN RESEARCH — RO-16: C4, C5, C7. Missing: C4: uniform certification of  $\|D_m E_m - I\|$  and end-to-end confirmation; C5: construction of  $R_{\text{eff}}$ , isolation of a non-trivial fixed point and certification by intervals/radii; C7: repetition under at least two admissible, non-conjugate regulators and comparison of the invariants. Falsifier F-G0-5: the only certifiable fixed point is trivial, ordered or outside the domain. Reference: Core 1.0 §3, §11 (steps 7–8); V346 p. 1918.

## II.12 — The confrontation contract

This chapter contains no new object. It fixes the contract of the deferred book.

Rule 12.1 (freeze). The complete MASTER V347 will be hashed (SHA-256) and dated before any new drafting of the confrontation. The V347.x armature versions remain auditable pre-freezes. After the freeze of MASTER V347, no correspondence found afterwards may silently modify a statement: any substantial modification will create V348 or an explicitly superseding revision.

Rule 12.2 (verdicts). For each object X of Books I-II, the confrontation asks the single question “what does X resemble?” and admits only four answers: IDENTITY [IDENTITÉ] (same object, same law), COMPATIBILITY [COMPATIBILITÉ] (no contradiction, no identity established), ANALOGY [ANALOGIE] (resemblance of structure without a common law), FAILURE [ÉCHEC] (contradiction). An ANALOGY can never be promoted to IDENTITY by resemblance of spectrum or by a potential approximately in  $1/r$  (Core 1.0 §7).

Rule 12.3 (current state). As of V347, all physical gates are at 0/7; the historical cosmological correspondences are [R]/[FAIL] and excluded; no label “quantum-gravity candidate” or “credible TOE candidate” may be used before the thresholds of Core 1.0 §7–§8 and §10 (current level: 1, reproducible mathematical model).

Rule 12.4 (honesty of the freeze). The freeze is retrospective: the narrative of Book I was recorded after 1918 pages of confrontation with reality. V347 therefore does not claim the virtue of a preregistration, but a strict separation of registers and a verifiable provenance. That is what the apparatus gives, and nothing more.

Correspondence with I.12. Residue: ABSENT [ABSENT — absent] by construction.

## II.13 — Without a twin

Chapter I.13 has no formal twin, by decision. It is in the opening register, and V347 refuses to fabricate a mathematical object for it in order to “pretend” (V346 p. 166, methodological remark on intentionally empty sections). The only thing Book II can say

about it is this: the observable proposed in the corpus for reflexivity,  
 $O(C) = S_{\text{sim}}(C) / (1 + D(C))$  (V346 p. 168), is a function defined on  $\Omega$ , measurable,  
and nothing more is asserted about it.

## BOOK III — THE REGISTER OF OPEN RESEARCH PROBLEMS

V347.1 consolidates 27 open research problems. They are separated into three bands: A = G0/continuum path; B = integrity of the causal mirror; C = foundations or claim bounds.

### III.1 — Order of work

#### Band A — G0 / continuum

| RO    | Object and missing piece                                      | Closure / falsifier | Ledger           |
|-------|---------------------------------------------------------------|---------------------|------------------|
| RO-12 | Confirmatory C2-R: independent replication and sizes held     | F-G0-1              | C2R              |
| RO-13 | GEN-4 typical-set: uniform local margin on typical sector     | F-G0-2              | GEN4-TYPICAL     |
| RO-14 | MARKED ↔ CONNECTED: intertwiner in declared norm              | F-G0-3              | MARKED-CONNECTED |
| RO-15 | Cutoff/projective: convergence under removal of the regulator | F-G0-4              | CUTOFF-PROJ      |
| RO-16 | C4/C5/C7: defect, non-trivial fixed point, universality       | F-G0-5              | G0-C4/C5/C7      |
| RO-18 | Finite window → thermodynamic limit                           | claim bound         | CORE-STATUS      |

#### Band B — integrity of the causal mirror

| RO    | Object and missing piece                    | Closure / falsifier                         | Ledger         |
|-------|---------------------------------------------|---------------------------------------------|----------------|
| RO-20 | Define “persist” independently of $\pi > 0$ | robust, non-arbitrary criterion             | VIABILITY-DEF  |
| RO-21 | Global accessibility / irreducibility       | exact components or proof of irreducibility | IRREDUCIBILITY |
| RO-22 | Generational improvement                    | monotonicity/non-monotonicity test          | GENERATION     |

| RO    | Object and missing piece                 | Closure / falsifier                                                      | Ledger                |
|-------|------------------------------------------|--------------------------------------------------------------------------|-----------------------|
| RO-23 | General dissipative threshold            | positive physical dissipation before $\Omega^3$ kills the strong version | DISSIPATION-THRESHOLD |
| RO-24 | Closure generator                        | no compatible local closure kills the causal cascade                     | CLOSURE-GENERATOR     |
| RO-25 | Growth $\rightarrow$ Gibbs               | no compatible joint extension                                            | GROWTH-GIBBS-BRIDGE   |
| RO-08 | A_Gibbs $\rightarrow$ B                  | uncontrollable topology                                                  | LAYER-BRIDGE          |
| RO-26 | Global twin state space                  | dynamics/RG incompatibility                                              | TWIN-STATE-SPACE      |
| RO-06 | $Q_{\text{samp}} \neq Q_{\text{phys}}$   | no admissible physical generator                                         | QTIME                 |
| RO-05 | Lineage $\sigma$ on $T^3$                | incompatibility with quotient/Pachner                                    | LINEAGE-T3            |
| RO-10 | Reduction operation $\mathcal{R}$ (bath) | dynamical incompatibility                                                | BATH-OP               |
| RO-09 | Lineage-structure coupling               | odd term forced to zero                                                  | LINEAGE-COUPLING      |
| RO-17 | Dissipation/rigidity on $T^3$            | non-reproduction under canonical action                                  | DISSIP-T3             |
| RO-07 | $t_1/t_2$ articulation                   | historical diagnostics not reproduced                                    | TIME-ARTICULATION     |
| RO-27 | Local entropic bookkeeping               | absence of coherent local production                                     | ENTROPY-BOOKKEEPING   |

## Band C — foundations and claim bounds

| RO    | Object and missing piece                                               | Closure / falsifier                                                      | Ledger            |
|-------|------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------|
| RO-02 | Nucleation $\Omega^0 \rightarrow \Omega^1$ + boundary convention for D | robust protocol or negative bound                                        | NUCLEATION        |
| RO-04 | Two simultaneous simplices                                             | complete reduction to existing operations or new operation made explicit | DOUBLE-SPLIT      |
| RO-03 | Derivation of $\sigma_0$                                               | closure as an axiom if obstruction from $\Omega^0$                       | SIGMA0-DERIVATION |
| RO-11 | Ratio $\eta$                                                           | forbidden before formal bath and Core derivation                         | COSMO-OLD         |
| RO-19 | Imposed topology                                                       | claim bound                                                              | CORE-STATUS       |

| RO    | Object and missing piece | Closure / falsifier | Ledger |
|-------|--------------------------|---------------------|--------|
| RO-01 | First movement           | out of frame by A2  | —      |

**Band A** directly blocks G0. **Band B** decides whether the narrative and the formalism can become a single causal theory. **Band C** contains the assumed origins and the limits of scope.

### III.2 — What an external contribution must provide

Every contribution must name the RO it attacks and provide, according to the nature of the claim: code and environment, data or deterministic generator, seeds, SHA-256 hash, negative test, a second independent engine for anything confirmatory, DEV/CONFIRMATORY/HELD-OUT separation, and a supersession report. A contribution that turns CARRIED into CORRECTED is a complete scientific result.

### III.3 — What V347.1 allows one to say

Here is the candidate as a **falsifiable causal specification**. It is not a demonstrated TOE. The work is localised in 27 open research problems: six on G0/continuum, fifteen on the integrity of the causal mirror, six foundational or of scope. Nothing from Book I is promoted to Book II without a corresponding object; nothing retracted is resurrected. The new corrections —  $\mu$ -only no-go,  $\partial\Delta^2/\Delta^2$  distinction, non-binary viability, non-general dissipative threshold, growth/Gibbs separation — reduce certain claims but make the programme harder to circumvent and therefore more scientific.

## SIFTING PLAN [PLAN DE DÉPOUILLEMENT] — from the armature to the complete corpus

The present document remains an armature. It is not honest to reduce V346 to 40 pages; it is just as unhelpful to copy out 1918 pages without hierarchy. The protocol keeps in extenso the proofs, data and retractions that modify a residue; it leaves the logs, duplicates and machine verbatim by hash/reference.

### P.1 — What V346 actually contains

| V346 pages | Content                                              | Volume | V347 destination       | Form                                 |
|------------|------------------------------------------------------|--------|------------------------|--------------------------------------|
| 1-155      | Version log<br>V107→V346                             | 155    | Provenance             | provenance lines,<br>not transported |
| 156-162    | Axioms A0-A6                                         | 7      | II.0, II.1, II.10      | in extenso                           |
| 163-216    | Annex book<br>ch. 1-6, $\Omega^0\rightarrow\Omega^3$ | 53     | II.1-II.6; Book I      | in extenso,<br>statuses reviewed     |
| 217-300    | Annex book<br>ch. 7-19                               | 83     | deferred book;<br>I.13 | confrontation,<br>many [R]           |

| <b>V346 pages</b> | <b>Content</b>                                   | <b>Volume</b> | <b>V347 destination</b>              | <b>Form</b>                          |
|-------------------|--------------------------------------------------|---------------|--------------------------------------|--------------------------------------|
| 300-319           | External audit / d=3 singularities               | 19            | provenance; II.6 [ARCH]              | summary + cross-reference            |
| 319-374           | Code and raw results                             | 55            | reproduction annex                   | hash/reference                       |
| 374-538           | Assessments V116-V128R                           | 164           | provenance; II.4, II.6, II.10 [ARCH] | selective extraction                 |
| 538-604           | Thematic developments                            | 66            | II.4, II.6 [ARCH]; deferred book     | selective extraction                 |
| 604-618           | Annexes A1/A2, proofs                            | 14            | II.4, II.10                          | in extenso, re-audited               |
| 618-714           | Unification, gravity, chirality, phase           | 96            | II.9, II.10 [ARCH]; deferred book    | extraction; physics deferred         |
| 714-800           | Calibration, $t_1$ , old cosmology               | 86            | II.4; deferred book                  | extraction; cosmology provenance/[R] |
| 800-930           | JSON and verbatim demonstrations                 | 130           | reproduction annex                   | by reference                         |
| 930-1141          | Symmetries, antimatter, Dirac-Kähler, twins, bit | 211           | II.2, II.4, II.7-II.9 [ARCH]         | in extenso for relevant parts        |
| 1141-1184         | Companion book / "why"                           | 43            | Book I, I.13, II.6                   | narrative sources                    |
| 1184-1265         | Selection, +infme [+tiny], sign, viability       | 81            | II.8-II.10                           | in extenso for relevant core         |
| 1265-1609         | SUP-1, ergodicity, limit, fields/fermions        | 344           | II.10, RO-18; deferred book          | selective extraction                 |
| 1609-1664         | BLOB-TOE $\beta$ 1.1, G0-G6, Q-TIME              | 55            | II.0, II.4, II.12, III               | in extenso for contracts             |
| 1664-1729         | Canonical $T^3$ sector                           | 65            | II.6, II.10, II.11                   | in extenso                           |
| 1729-1907         | G0 mechanics + outputs                           | 178           | II.11 + reproduction annex           | theory in extenso; verbatim by hash  |
| 1907-1918         | V345-V346, breathing, R-TRUE, GEN-4              | 12            | II.10, II.11                         | in extenso                           |

The removal of logs, duplicates and machine outputs is not a summary of the substance: the proofs, data, retractions and dependencies that change a residue are kept; the reproducible verbatim remains addressable by hash.

## P.2 — Target volume of the complete MASTER V347

| Book               | Content                                               | Target volume     |
|--------------------|-------------------------------------------------------|-------------------|
| Canonical armature | rules, mirror, residues, RO, glossary                 | 40–55 pp.         |
| Book I             | developed cosmogony + provenance                      | 60–90 pp.         |
| Book II            | 13 formal chapters, proofs, certificates, retractions | 380–500 pp.       |
| Book III           | 27 complete RO sheets                                 | 50–80 pp.         |
| Deferred book      | physical confrontation under 4 verdicts               | 250–350 pp.       |
| Reproduction annex | code, JSON, outputs                                   | by hash/reference |
| Editorial total    | excluding machine verbatim                            | 780–1,075 pp.     |

The target increases slightly relative to V347.0 because V347.1 adds the A\_gen/A\_Gibbs strata, the closure/topology no-gos and five new ROs; it remains far below V346 because repeated provenance and raw outputs are not reprinted.

## P.3 — Revised production protocol

The sifting no longer follows the order of certainty of the results, but the order of causal risk: the links that could invalidate the mirror are re-audited before being developed over hundreds of pages.

1. II.3 + II.5 — mitosis,  $\mu$ -only no-go, closure, boundary/filling; create the RO-24 sheet.
2. II.4 + II.6 —  $t_0/t_1/t_2$ , KL sign, dissipative threshold, distinct thresholds of S\_sim; create RO-23.
3. II.10 — viability, persistence, accessibility, C2-R, entropic bookkeeping; RO-20/21/27.
4. II.0 — formalise the A\_gen/A\_Gibbs/B strata and the RO-25/RO-08 bridges; integrate the B\_full/AUT-LIFT repair.
5. II.11 — transport in extenso the R-TRUE/full-link/pull-back/MARKED branch, currently the most certified.
6. II.2 + II.7 + II.9 — lineage, twin space, chirality; RO-26.
7. II.8 — bath: document exactly the formal void and RO-10/11.
8. II.1 — initial object, boundary of D/S\_sim and RO-01/02.
9. Book I developed, keeping the historical provenance and without feedback from the solutions found in Book II.
10. Book III as complete sheets, and only then the deferred book of confrontation.

Each delivered chapter comprises: in extenso text, claim IDs, V346 cross-references, dependencies, retractions, parameters and their provenance, reproduction package, and a log line.

**Safety rule.** No historical [T] enters without direct recomputation. V347.1 has already found three classes of errors that had passed through hundreds of pages: sign of the free

energy,  $\mu$  unable to close,  $\partial\Delta^2/\Delta^2$  confusion. The sifting must assume that others exist until proven otherwise.

## Minimal glossary

- **A\_gen**: growth/genealogy stratum with variable  $N$ ; carries  $\mu$  and  $\sigma$ .
- **A\_Gibbs**: historical [ARCH] engine with fixed  $N$ , Rips/flag graph,  $S_{\text{sim}}$  and edge-swap.
- **B\_shell / B\_full**:  $T^3$  sectors with fixed / variable  $N_0$ .
- **Flag (clique) complex**: every clique of the 1-skeleton is filled in as a simplex; in particular  $K_3$  gives  $\Delta^2$ .
- $\partial\Delta^2 / \Delta^2$ : triangular boundary ( $S^1$  cycle) / filled triangle (contractible disc).
- **Mitosis  $\mu$** : addition of a vertex and of a single edge towards an existing vertex; on its own, it preserves the tree structure.
- **Closure**: still-open mechanism that connects existing vertices and allows leaving the tree structure.
- **Lineage  $\sigma$** : inherited  $\pm 1$  label;  $\sigma_0$  is posited.
- **$t_0, t_1, t_2$** : absence of order; genealogical counter; accumulation of  $D$  along the sampler.
- **D(C)**: coefficient of variation of the degrees, defined when the mean degree is strictly positive.
- **$\mathcal{L}[\nu]$** :  $\tau_{\text{KL}}(\nu \parallel \pi)$ , correct Lyapunov function of the historical sampler.
- **R-TRUE**: edge contraction satisfying the full link condition.
- **Pull-back /  $\Delta m$** : canonical transport of a coarse domain and variation of its tetrahedron count.
- **Margin  $\omega$** : finite local bound of the MARKED generator.
- **Residue**: CARRIED [PORTÉ], WEAKENED [AFFAIBLI], CORRECTED [CORRIGÉ], ABSENT.

## Log of V347.1 corrections before the armature freeze

V347.1 keeps the corrections from the two audits of V347.0 and adds a third, causal audit.

### Corrections inherited from V347.0

- $K_3 = 15/32$  relocated as a property of the Rips substrate [T][ARCH].
- $dt_2/dt = D$  requalified as a definitional identity.
- reservation on the statistical independence of the chiral  $P = 0.516$ .
- sign of the free energy corrected by  $\mathcal{L} = \tau_{\text{KL}}$ .
- historical  $A_0$  retracted as a binary filter;  $A_0' + RO-20$ .
- $B_{\text{shell}}/B_{\text{full}}$  distinguished.
- “simplicial carrier” locked.
- neutrality of the twins made conditional.
- $\tau_c$  withdrawn from any universal pretension.
- global accessibility RO-21; generational improvement RO-22.

### Third causal audit — V347.1

1.  **$\mu$ -only NO-GO [T].**  $\mu$  adds a leaf; repeated on its own, it remains a tree. The cascade towards  $\Omega^2/\Omega^3$  requires a closure mechanism: RO-24.
2.  **$\partial\Delta^2/\Delta^2$  correction [T]/[R].** The triangular boundary carries the loop; the filled 2-simplex is contractible. In flag/Rips,  $K_3$  is filled and does not create  $H_1$ .
3. **Universal dissipative threshold [R].** The corpus does not exclude  $D>0$  as early as irregular trees, nor in dimension 2; with  $t_2 = \Sigma D\Delta t$ , the former “no arrow at  $\Omega^1$ ” is general only in the regular subclass. The  $\Omega^3$  threshold becomes conditional. RO-23.
4. **Threshold of  $S_{\text{sim}}$  [R].** An irregular tree can have  $S_{\text{sim}} = -\beta D \neq 0$ . The first triangle is only the first ignition of the term  $\log(1+n_{\text{tri}})$ , not of the whole functional.
5.  **$\Omega^0$  frontier.**  $D$  is undefined when the mean degree is zero; no convention  $D(\Omega^0) = 0$  is added silently. RO-02 is broadened.
6. **A\_gen/A\_Gibbs separation.** Growth by mitosis and the Gibbs sampler at fixed  $N$  are two distinct mechanisms; their bridge becomes RO-25.
7. **B\_full.** The AUT-LIFT/full Pachner tension is requalified as a largely acquired operational repair, with the formal certificate remaining under RO-16.
8. **Two universes.**  $\sigma$  on a graph does not construct two twin state spaces; RO-26.
9. **Viability.** “Everything appears / not everything persists” is decomposed into exploration, accessibility, weighting and persistence criterion; residue of II.10 lowered to WEAKENED.
10. **Thermodynamic price.** The decrease of KL is not a proof of entropy rejected into an environment; RO-27.
11. **Internal rules.** The prohibition of the word “time” in Book II and the rule “every term defined before use” were themselves violated; they are replaced by rules of physical vocabulary and of explicit dependencies.
12. **Count of controls.** 10.4 comprises five controls, not four.

These corrections add no physics. They remove shortcuts and expose the operations that are actually missing.

## Provenance, versioning and freeze

**Sources.** BLOB PRINCIPLE — CORE 1.0 and its Claim Ledger for the current statuses; Master V346 (1918 pages) as provenance archive and source of the cited pages; V347.0 as the first armature.

### Version rule.

- V347.0: first mirrored armature.
- V347.1: canonical armature after the third causal audit.
- V347.x: armature and sifting iterations as long as Books I-III are not complete.
- MASTER V347: name reserved for the complete corpus, once Books I and II are frozen and hashed.
- V348: first substantial modification after the freeze of MASTER V347.

**What V347 does not claim.** It derives neither  $T^3$  from  $\Omega^0$ , nor a physical clock from Gibbs, nor the bath, nor two complete universes, nor a generational improvement, nor a continuum limit, nor an observed physical theory. These absences are ROs, not rhetorical blanks.

***Corde de rappel*** (the author's safety-line principle; literally "abseil rope"). Nothing beyond what the corpus carries. No object invented to save a chapter. Every correction must be able to lower the programme's score. The depth of the Blob is preserved; only the ambiguity between intuition, definition, proof and identification is forbidden.